2 **(a)** The waves have a constant phase difference.

(not accepted: same amplitude, wavelength, velocity)

(b)
$$x = \frac{\lambda D}{a}$$
$$= \frac{530 \times 10^{-9} (2.7)}{0.85 \times 10^{-3}}$$
$$= 1.6835 \times 10^{-3} = 1.7 \times 10^{-3} \text{ m}$$

(c) **(i)** Less light passes through the narrower slits resulting in dimmer bright fringes. There is no change in the fringe separation as slit width is unchanged.

(note: 'decrease in contrast' is ambiguous. It must be clear that bright fringes are dimmer only. Dark fringes remain dark.)

(ii) As the screen is rotated, the distance between different parts of the screen to the double slit will not be the same.

The fringe separations will be shorter when the screen is nearer to the double slit, and longer on the part of the screen further away (fringe separation is proportional to slits-to-screen distance).

The fringes when the screen is nearer to the double slits are brighter and the fringes when the screen is further away are dimmer (light is spread over a larger area when the screen is further away).